

Part 1: Theoretical Understanding (30%)

1. Short Answer Questions

Q1: Define algorithmic bias and provide two examples of how it manifests in AI systems.

Algorithmic bias is when an AI system makes systematic errors that unfairly favor one group over another. It is learned from data that is flawed, incomplete, or reflects historical prejudice.

- **Example 1: Facial Recognition.** Many systems are trained on datasets with mostly light-skinned male faces. As a result, they have higher error rates when identifying darker-skinned individuals, especially women.
- **Example 2: Loan Approval AI.** An AI trained on historical loan data might learn that applicants from certain neighborhoods (which are proxies for race or wealth) are higher risk. It then unfairly denies loans to qualified people from those areas.

Q2: Explain the difference between transparency and explainability in AI. Why are both important?

- **Transparency** is about seeing *how* the AI system was built. (e.g., "What data was it trained on?" "What is the source code?").
- **Explainability** is about understanding *why* the AI made a *specific decision*. (e.g., "Why was *my* loan application denied?").

Both are important for trust and accountability. Transparency proves the system was built fairly. Explainability gives a user the right to an answer and a way to appeal a decision.

Q3: How does GDPR (General Data Protection Regulation) impact AI development in the EU?

GDPR forces AI teams to adopt "privacy by design." Developers cannot just use any data they find. They must have a legal basis (like user consent) to collect data, must not collect more than they need, and must secure it. Most importantly, GDPR includes a "right to explanation," which makes it risky to use "black box" models that cannot explain their decisions.

2. Ethical Principles Matching

- **B) Non-maleficence:** Ensuring AI does not harm individuals or society.
- **C) Autonomy:** Respecting users' right to control their data and decisions.
- **D) Sustainability:** Designing AI to be environmentally friendly.
- **A) Justice:** Fair distribution of AI benefits and risks.

Part 2: Case Study Analysis (40%)

Case 1: Biased Hiring Tool

- **Source of Bias:** The core problem was the **training data**. The AI was trained on 10 years of Amazon's past resumes. Since the tech industry was male-dominated, the AI learned to prefer male candidates and penalize resumes containing words like "women's" (e.g., "women's chess club").
- **Three Fixes:**
 1. **Data Fix:** Re-sample and balance the training data to include an equal number of successful male and female candidates.
 2. **Feature Fix:** Audit and remove features (words or phrases) that are strong proxies for gender (like "women's" or all-female college names).
 3. **Model Fix:** Use a "fairness-aware" model. Add a constraint that forces the model to select male and female applicants at a similar rate, optimizing for both fairness and skill.
- **Metrics to Evaluate Fairness:**
 1. **Demographic Parity:** Check that the *percentage* of male and female applicants who are *selected* is roughly the same.
 2. **Equal Opportunity:** Check that *qualified* male and *qualified* female applicants are *selected* at the same rate.
 3. **Regular Audits:** Continuously log and review the live model's decisions to catch new biases that may appear over time.

Case 2: Facial Recognition in Policing

- **Ethical Risks:**
 1. **Wrongful Arrests:** The high error rate for minorities creates a direct risk of arresting innocent people. This can ruin lives and has already happened.
 2. **Mass Surveillance & Privacy Violation:** It allows the government to track everyone in public places without their consent, which has a chilling effect on free speech and the right to protest.
 3. **Reinforcing Bias:** If police use this tool more often in minority neighborhoods, the system will (correctly or incorrectly) "find" more suspects there. This creates a data feedback loop that justifies even more surveillance in those same communities.
- **Policies for Responsible Deployment:**
 1. **Ban on Live Monitoring:** Ban the use of facial recognition for live, real-time surveillance in public spaces.
 2. **Strict Human-in-the-Loop:** An AI match can *never* be the only reason for an arrest. It can only be a "tip" that an officer must fully investigate and confirm with separate evidence.
 3. **Independent, Public Audits:** Before any police department can buy a system, it must be independently tested for accuracy across all races and genders, and the results must be made public.